



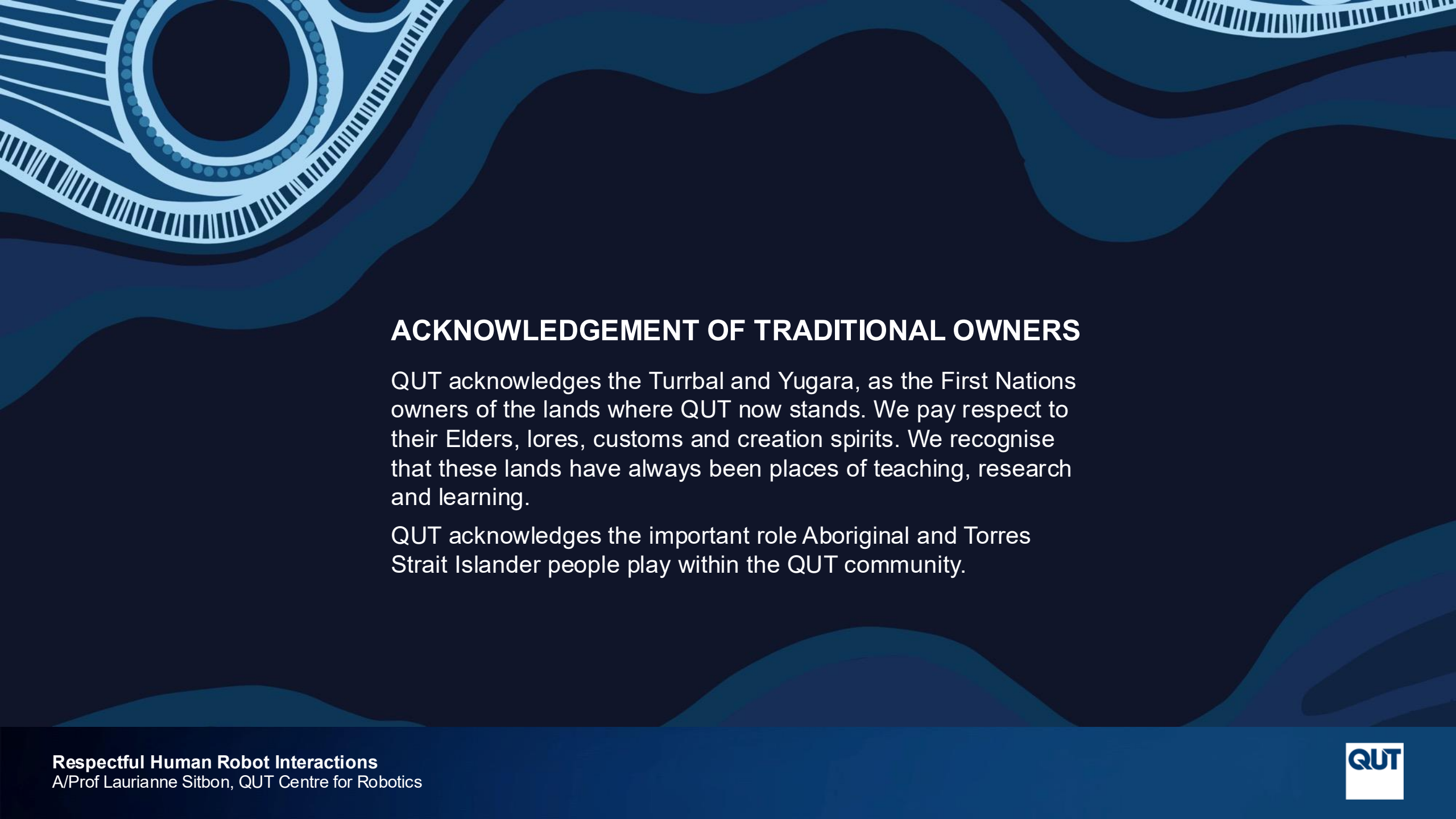
Centre for
Robotics



Respectful Human- Robot Interaction

Associate Professor Laurianne Sitbon
QUT Centre for Robotics

AuSRoS 2025



ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

My Research journey

Information Retrieval

Robustness

(Latent) Semantics
Concept extraction
(clinical)

Service Learning

Volunteering students'
learning opportunities

Endeavour Foundation
(people with intellectual
disability)

HCI

Co-design

Accessibility

Human Robot
Interaction

Affiliations

- QUT School of Computer Science
 - Human-Computer Interaction
- QUT Centre for Robotics
 - Human-Robot Interaction and Collaboration
 - Inclusive Robotics (social robots, AV, rehabilitation, navigation assistance)
- QUT Disability Research Network
 - Creative Industries, Engineering, Social Work, Business, Architecture, Education
- QUT Centre for Justice
 - Technologies of Justice

Overview

Brief overview of Human Robot Interaction

Why respectful interactions matters

Social Distancing

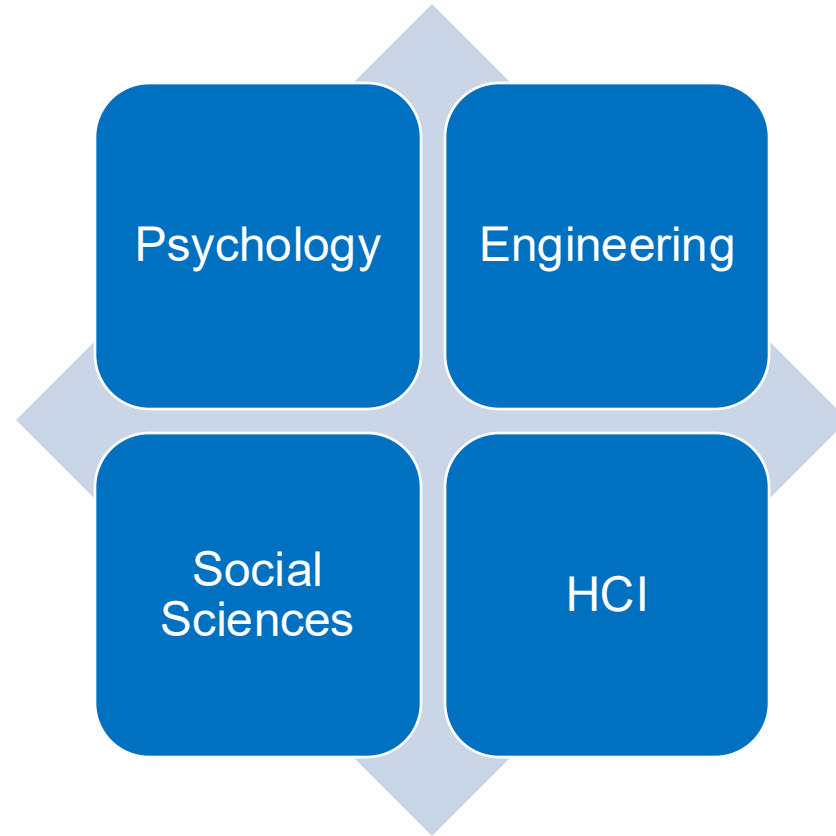
Cognitively-aware interaction and communication

Qualitative and Design Approaches to HRI

Brief Overview of HRI

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Disciplines



Modeling Behaviour

Theory of Mind

Predict human behaviour by modeling mental states

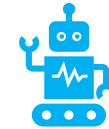
- beliefs
- intentions
- goals



Mental Models

Predict robot behaviour through experiential learning

- Structure and relationships
- Analogical reasoning
- “intuition”



Effectiveness Studies

- Defined by (standardized) measurable outcomes
 - Task: speed, accuracy
 - Learning: recall, problem solving
 - Cognitive load: situational awareness, eye movements, stress (physiological)
 - Engagement: usability, trust, enjoyment
- Generally, lab-based studies
 - Controlled variations, standardized context
 - Control groups

Human-Robot Relationships

- Applications
- Collocation
- Control

Application area	Remote/ Proximate	Role	Example
Search and rescue	Remote	Human is supervisor or operator	Remotely operated search robots
	Proximate	Human and robot are peers	Robot supports unstable structures
Assistive robotics	Proximate	Human and robot are peers, or robot is tool	Assistance for the blind, and therapy for the elderly
	Proximate	Robot is mentor	Social interaction for autistic children
Military and police	Remote	Human is supervisor	Reconnaissance, de-mining
	Remote or Proximate	Human and robot are peers	Patrol support
	Remote	Human is information consumer	Commander using reconnaissance information
Edutainment	Proximate	Robot is mentor	Robotic classroom assistant
		Robot is mentor	Robotic museum tour guide
Space	Remote	Robot is peer	Social companion
	Remote	Human is supervisor or operator	Remote science and exploration
Home and industry	Proximate	Human and robot are peers	Robotic astronaut assistant
	Proximate	Human and robot are peers	Robotic companion
	Proximate	Human is supervisor	Robotic vacuum
	Remote	Human is supervisor	Robot construction

Assistive Robotics

- Socially Assistive Robotics
 - Older adults companionship
 - Cognitive support for neurodiverse people
 - Healthy habits coach
- Rehabilitation
 - Orthotics/prosthetics
- Mobility support
 - Home care
 - Robotic wheelchairs



Therapeutic baby seal robot PARO. Photo credit: S. Šabanović

Social Robots

- Anthropomorphic features
 - Eyes, movement, arms, legs, voice, function (!)
- Anthropomorphism
 - Added expectations: social, communication, behaviour
 - Supports trust and acceptance
 - Cognitive bias (trusting humans -> trusting human-like tech)
- Social Features
 - Communication: voice, screens
 - Purpose

Humanoid Robots

- Humanoid features
 - Human height – can work in human spaces
 - Walking – can move through steps, stairs, etc.
 - Proactive and responsive behaviours
- Expectations
 - Social features
 - Directed visual capabilities (watch your back!)
 - Voice Recognition

Respect!

What does “respectful” even mean?

Respectful Robots in Physical and Communication Spaces

- Safety risks
 - Physical safety
 - Impacts on cognitive load
- Annoyance risks
 - Disuse
 - Misuse
- Side effects
 - Transformed human/human interactions

Trust

Measuring Trust

- Human-Computer Trust Scale (HCTM)
 - competence, benevolence, and risk perception
- Trust Perception Scale-HRI (TPS-HRI)

Calibrating Trust

- Commensurate with task criticality

Other Trust Factors

- Reliability, Capability
- Appearance
- Transparency

Social Distancing

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Avoidance

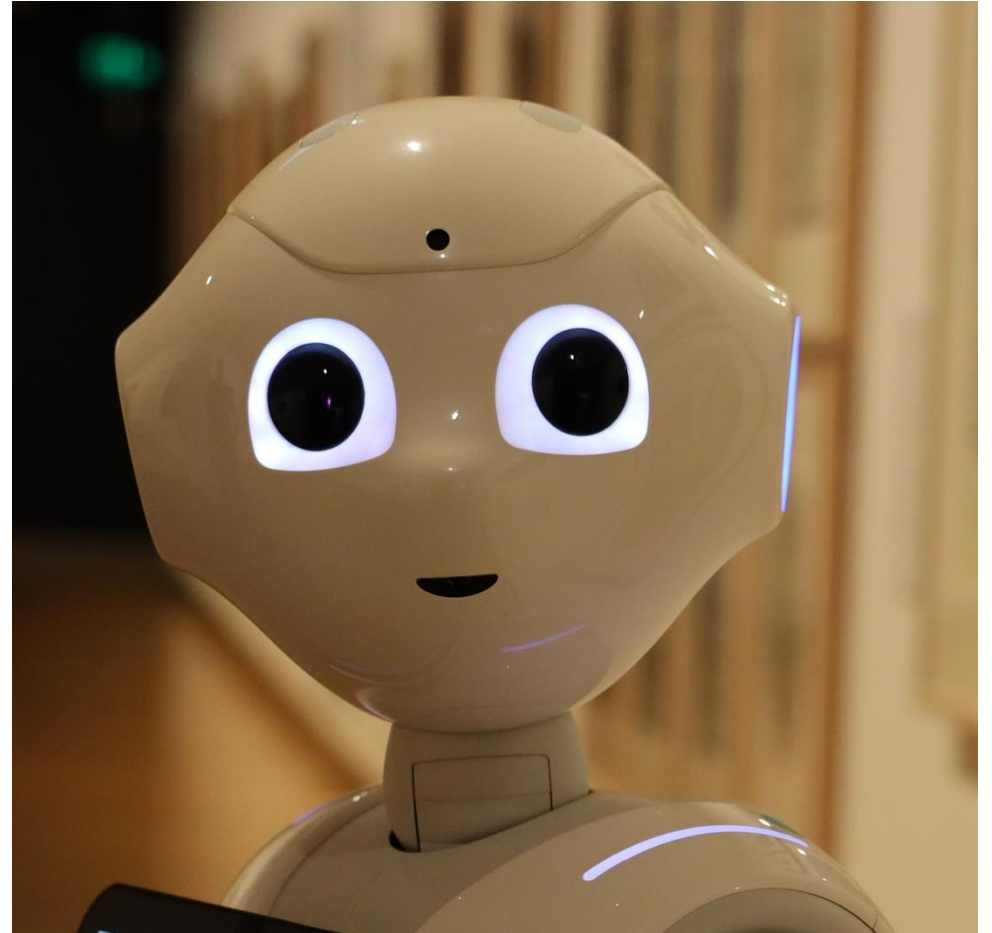
- Collision avoidance mechanisms are established and necessary for **safety**
- Respectful interactions need to consider
 - Acceptable Distance (Proxemics)
 - context/culture dependent
 - Avoidance timing



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Users/people as input devices

- Intention prediction
 - Body (natural) movement, eye tracking, scene interpretation
 - Proactive behaviour
 - Invasion of “private space”?
- Gesture-based interaction
 - Intentional movement
 - Responsive behaviour
 - Always on...
 - Invasion of “private space”?



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Predictable Behaviour

- Socially-aware navigation
 - Corridor classic
 - Side by side navigation
- Frameworks for Reciprocal Avoidance
 - Relational positions: both are moving and making decisions dynamically



Attempted Getaway | Traffic madness on the Place de l'Etoile... | Jez Elliott | Flickr

Physical assistance

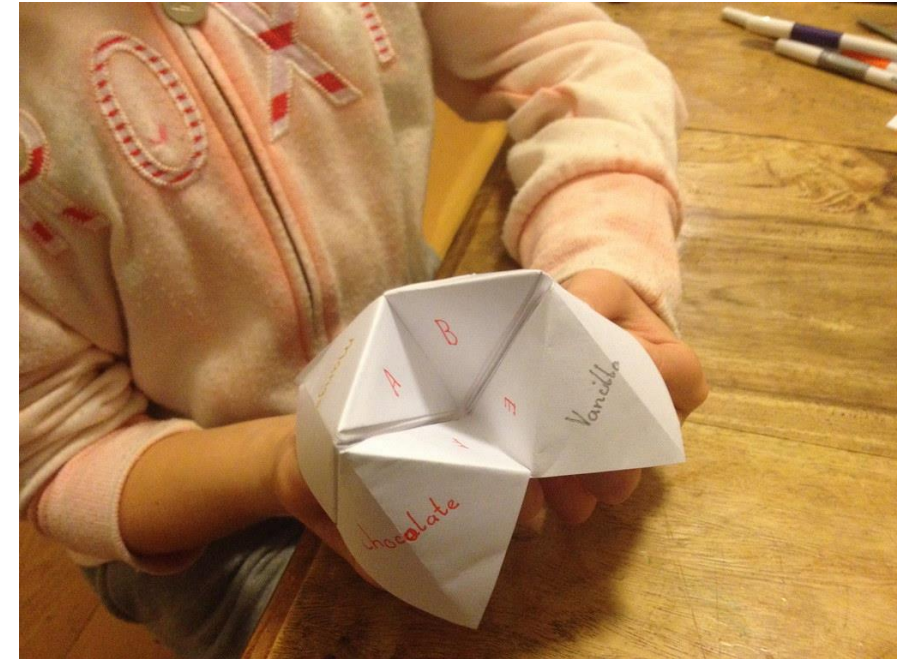
- Dignity Preservation for Assistive Robots
 - Let people do as much as they can/want
- Co-manipulation
 - Negotiating “movement leadership”

Cognitively-aware interaction and communication

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Invasion of Communication Spaces

- Communication Spaces
 - Limited “space” (time)
 - Busy “space” (attention)
- Proactive Communication
 - Interrupting tasks
 - Interrupting people
 - Causing distraction



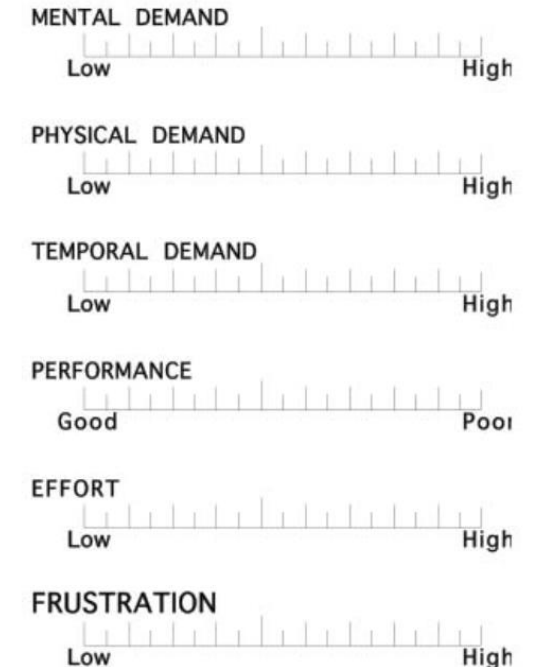
Source: [how to make a chatterbox kids IMG 0969 | Nicole | Flickr](#)

Navigating Communication Spaces

- Explainable AI
 - Too much information ?
 - Needs Prioritisation
- Maintaining situational awareness
- Measuring Cognitive Load

RATING SCALE DEFINITIONS		
Title	Endpoints	Descriptions
MENTAL DEMAND	Low/High	How much mental and perceptual activity was required (e.g., thinking, deciding, calculating, remembering, looking, searching, etc.)? Was the task easy or demanding, simple or complex, exacting or forgiving?
PHYSICAL DEMAND	Low/High	How much physical activity was required (e.g., pushing, pulling, turning, controlling, activating, etc.)? Was the task easy or demanding, slow or brisk, slack or strenuous, restful or laborious?
TEMPORAL DEMAND	Low/High	How much time pressure did you feel due to the rate or pace at which the tasks or task elements occurred? Was the pace slow and leisurely or rapid and frantic?
EFFORT	Low/High	How hard did you have to work (mentally and physically) to accomplish your level of performance?
PERFORMANCE	Good/Poor	How successful do you think you were in accomplishing the goals of the task set by the experimenter (or yourself)? How satisfied were you with your performance in accomplishing these goals?
FRUSTRATION LEVEL	Low/High	How insecure, discouraged, irritated, stressed and annoyed versus secure, gratified, content, relaxed and complacent did you feel during the task?

Task Load Index (NASA-TLX)



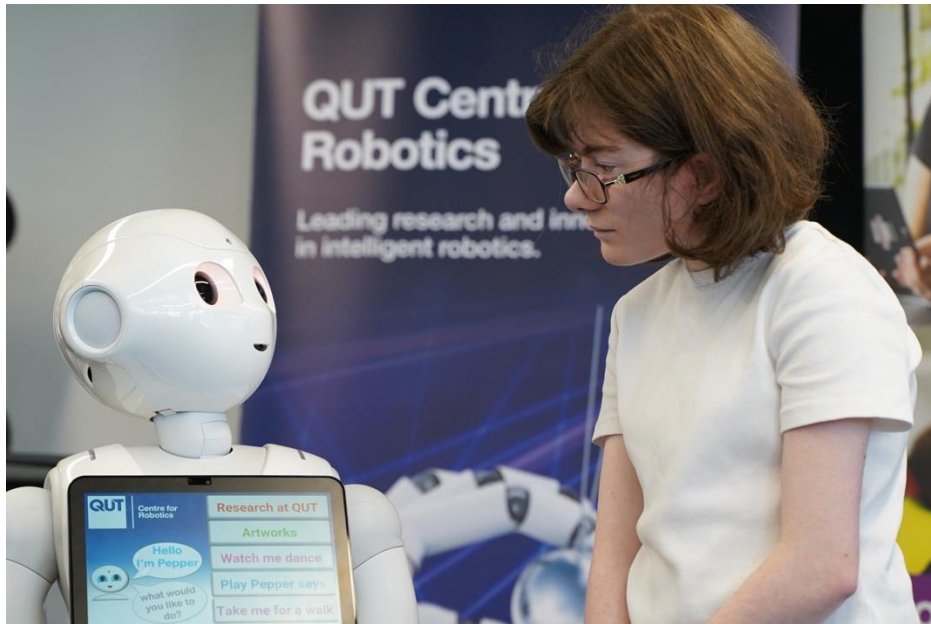
Challenging Environments

- External Cognitive Load Factors
 - Heavy Mental Workload
 - Time Pressure
 - Stress
- How can robots offload cognitive load?
 - What is worth offloading?



Stanford Report, Stanford's OceanOneK connects human's sight and touch to the deep sea, Taylor Kubota, Stanford Report, <https://news.stanford.edu/,2022>

Diverse Cognitive Abilities



"It helped me with my goal of wanting to become more independent and move out of home. I feel like Pepper can help me with moving out of home with cooking etc."

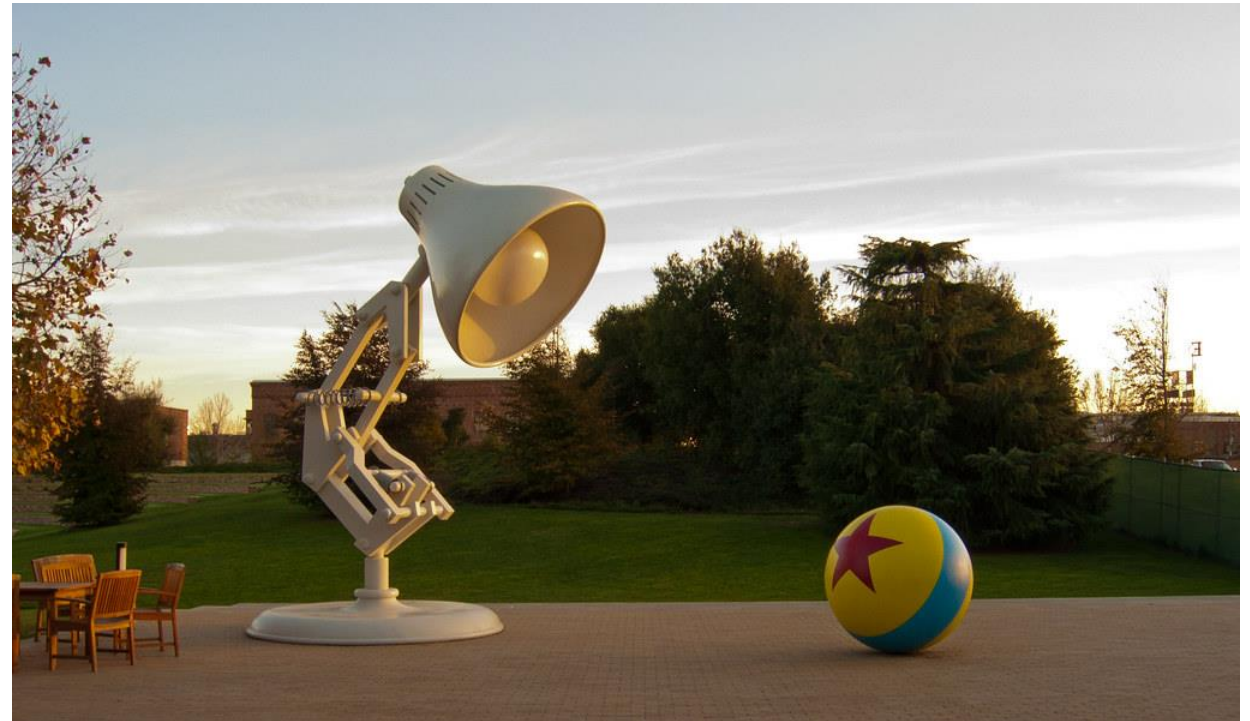
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Respectful Human Robot Interactions
A/Prof Laurianne Sitbon, QUT Centre for Robotics

Haidenhofer, Chloe, Sitbon, Laurianne, Beaumont, Chris, Hoogstrate, Maria, & Korte, Jessica (2024) From research participant to co-researcher: Chloe's story on co-designing inclusive technologies with people with intellectual disability. In Flatla, David, Hwang, Faustina, Guerreiro, Tiago, & Brewer, Robin (Eds.) *ASSETS '24*:

Robot Communication through Multimodal Channels

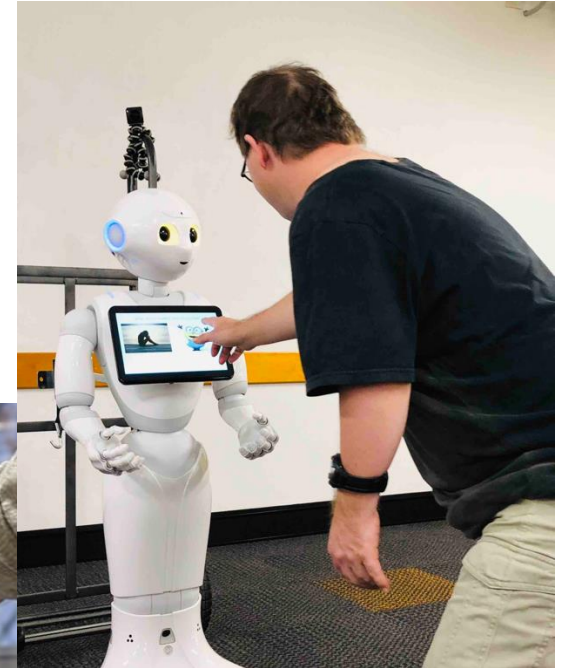
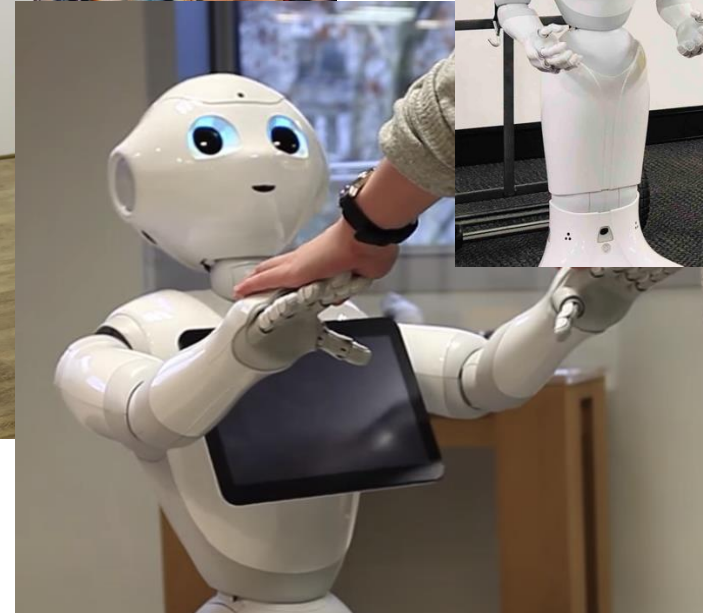
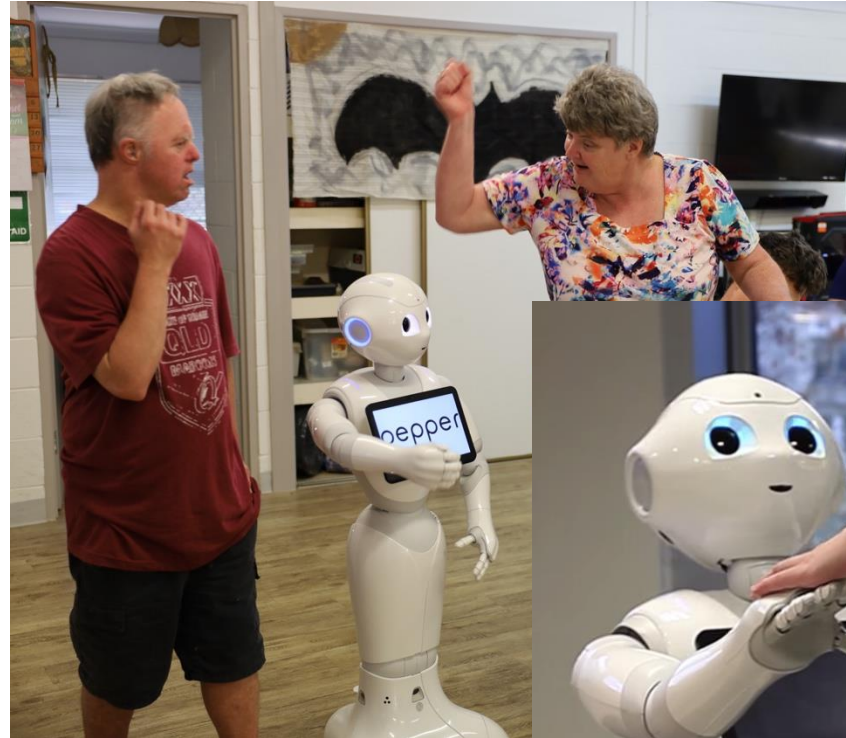
- Natural Language
 - Voice
 - Text
- Semantics of
 - Movement
 - Lights
 - Sounds
- Gaze



[Luxo Jr.](#) | [Jason Pratt](#) | [Flickr](#)

User expression through multiple modalities

- Movement
- Gestures
- Touch
- Buttons
- Speech
- Text



Qualitative and Design Approaches

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Value Sensitive Design

- “Safety by Design”
 - Built in, not bolted on
- Discover and characterize all stakeholders values
 - Utility (economics, value for money/time, etc.)
 - Usability (self-determination, learning curve, etc.)
 - Trust (privacy, safety, respect, etc.)
- Informs
 - Features
 - Algorithms
 - Strategies

Qualitative Research for Engineers



Attitude

Genuine interest in what people impacted by the tech have to say

Empowering people to have a voice throughout the process

Develop listening skills



Questions

From "does this work?" to "what is going on?"

From "how to solve someone's problem?" to "how to help someone solve the problem?"

From "what does industry want?" to "what do people want?"

From solving problems to exploring opportunities

Engineering Mindset:

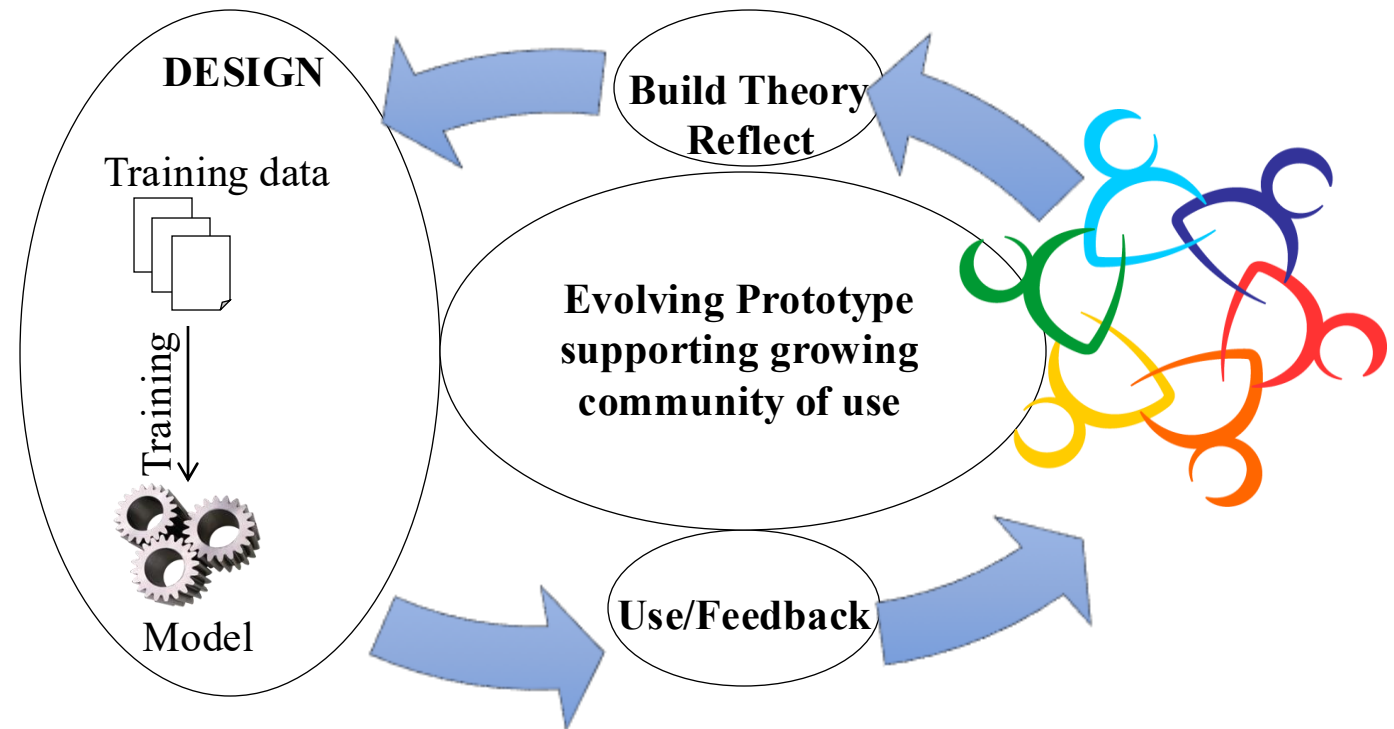
- Define a problem
- propose a solution to the problem,
- evaluate the solution, measure the outcomes.
- Perhaps people (research participants) help with measuring the outcomes by using the solution.

Qualitative Mindset:

- Explore the technology/society nexus with a community,
- identify opportunities,
- frame research,
- explore opportunities,
- consolidate knowledge,
- reframe research.

Reflective Agile Iterative Design (RAID)

- Codesign and Research Method at once
- Exploratory Prototypes
 - Technology Probes with a purpose
 - Not for evaluation, but for learning



Qualitative Methods

- Ethnography and observations
 - Detailed account of context
 - Exhibits people's preferences through actions
 - Time consuming
- Interviews and focus groups
 - Semi-structured: guided conversations
 - User Expectations, Values, Priorities, Comfort levels
 - Early stages
- Contextual Inquiry
 - Observations and Interviews while Using Robots

Research through co-design

- Form of participatory design
- Rich context for socio-technical interferences
- Communities drive the research agenda (pre-design, ideation, iterative prototyping, evaluation)
 - Power balances
 - Avoiding Disability Dongles



Codesign can be ...



Codesign workshops (post-it notes!)

Ideation, Agenda
Early designs



Field trials

Observations
Contextual interviews



Ongoing feedback loop

Emails, Miro Board, comments on Blog Posts
Diaries (text, pictures or video of technology in use)

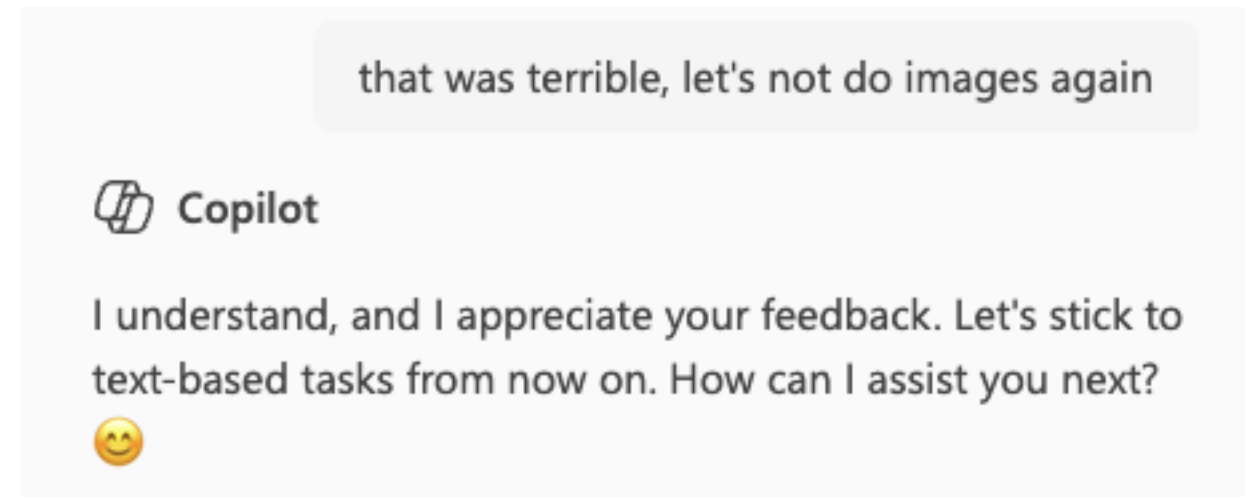


Creative Methods

Techshops
Arts and Craft

Design for “Mediocre AI”

- AI/ML can never yield perfect results
- Designing coping mechanisms for breakdowns
 - Confident errors break trust
 - Communicate uncertainty
 - Establish recovery strategies
- Possible Approaches
 - “Apologise”, take responsibility
 - Clarify uncertainty
 - Explain



Making sense of f-measures



We found a significant 2% improvement in accuracy (from 78 to 80%)



Our approach addresses the issue 90% of the time



The error rate is down 5%, only at 15%

Take Home Messages

- Engage early and often with HRI research relevant to your work
 - It can guide your tasks, features and algorithms!
- Engage early and often with the humans who will be around your robots
 - Qualitative approaches can give you great insights!
- Consider including an HRI studies in your PhD
 - HRI studies cannot "fail", whatever you try, there is always something to learn!
- Your robots/systems are only proxy for your own respect for people.